Precinct × vote-mode election-night model (TX Governor trial) · Young-Kyung Kim · October 18, 2026

This documents the model implemented in `scripts/model/tx_gov_trial.R`: what is estimated, how, and what it assumes. The design goal is a forecast of each candidate's share of all votes that updates as returns arrive by vote mode, with honest uncertainty under repeated looks. Section 1 fixes notation, 2–5 give the model and its estimation, 6–7 the prediction machinery, 8 the intervals, and 9 the assumptions and known gaps.

1. Setup and the composition

Index precincts $i = 1, \dots, N$ and vote modes $m \in \mathcal{M} = \{\text{mail}, \text{early}, \text{eday}\}$, with $M = |\mathcal{M}| = 3$. The unit of analysis is the *cell* (i, m). When a cell has reported we observe three counts $v^{\text{dem}}_{im}, v^{\text{rep}}_{im}, v^{\text{oth}}_{im}$, where oth collects every candidate outside the two major parties, including write-ins.

Rather than model the three counts directly, the model works with a factorisation of the composition:

$$\tau_{im} = v^{\text{dem}}_{im} + v^{\text{rep}}_{im} + v^{\text{oth}}_{im}, \qquad \text{tp}_{im} = \frac{v^{\text{oth}}_{im}}{\tau_{im}}, \qquad s_{im} = \frac{v^{\text{dem}}_{im}}{v^{\text{dem}}_{im} + v^{\text{rep}}_{im}}, \tag{1}$$

so τ is all votes cast in the cell, tp is the third-party share *of all* votes, and s is the Democratic share of the *two-party* vote. The map is invertible:

$$v^{\text{dem}}_{im} = \tau_{im}(1 - \text{tp}_{im})s_{im}, \qquad v^{\text{rep}}_{im} = \tau_{im}(1 - \text{tp}_{im})(1 - s_{im}), \qquad v^{\text{oth}}_{im} = \tau_{im}\text{tp}_{im}. \tag{2}$$

Because (1)–(2) is a bijection between $(\tau, \text{tp}, s) \in \mathbb{R}_+ \times (0,1)^2$ and $(v^{\text{dem}}, v^{\text{rep}}, v^{\text{oth}})$, nothing is lost relative to modelling the counts, and the three implied shares of all votes sum to one and lie in $[0, 1]$ *by construction* — no clamping and no post-hoc renormalisation.

This matters because the obvious alternative — one model per candidate, then divide by the row sum — is not innocuous. With two candidates the predictive residuals are drawn independently when they should be perfectly anticorrelated, and the division cancels part of the genuine noise; simulation against the implementation puts the resulting predictive standard deviation about 29% too small. Factorising avoids this and leaves the inter-candidate dependence to be estimated rather than induced.

2. The system of equations

Write $k \in \mathcal{K} = \{\tau, \text{tp}, s\}$ for the three outcomes. The precinct-level covariate vector x_i is deliberately small — with hundreds of races one cannot hand-curate a design per contest — and is given in Table 1. All of the continuous demographic terms are standardised, so the design mixes a log-count, two rates and four standardised scores; the g-prior of §3 is invariant to that rescaling, which is why no further normalisation is needed.

Note what is *absent*. There is no vote-mode covariate, because mode is a dimension of the unit rather than a regressor. There is no spatial basis. And the partisan baseline enters as a single term, `dem2`, derived from a vendor model calibrated to 2024 precinct returns — so the design carries one partisan signal, not the three that the NYC pipeline used. The modelled quantities are

$$y^{\tau}_{im} = \log(\tau_{im} + 0.1), \qquad y^{\text{tp}}_{im} = \text{logit}\left(\frac{v^{\text{oth}}_{im} + \frac{1}{2}}{\tau_{im} + 1}\right), \qquad y^{s}_{im} = \text{logit}\left(\frac{v^{\text{dem}}_{im} + \frac{1}{2}}{v^{\text{dem}}_{im} + v^{\text{rep}}_{im} + 1}\right). \tag{3}$$

The $+\frac{1}{2}$ terms are continuity corrections used *only for fitting*: a cell with zero third-party votes is common and would otherwise give $y^{\text{tp}} = -\infty$.

term	source	in which equation
`log_n`	log registrants, L2	turnout, third-party, share
`turnout22`	`vote22/n`, L2 (2022 midterm rate)	turnout, share
`turnout24`	`vote24/n`, L2	turnout, share
`age`	median age, voter file	turnout, third-party, share
`male`	male share of registrants	turnout, third-party, share
`pred.bla`	modelled Black share (BISG)	turnout, share
`pred.his`	modelled Hispanic share (BISG)	turnout, share
`dem2`	two-party Dem baseline, `pr_harris/(pr_harris+pr_trump)`	turnout, third-party, share
`tp_prior_raw`	`1−pr_harris−pr_trump`	third-party only
county indicators	one column per county	all three (penalised; §4)

Table 1: The covariate vector x_i. Turnout and share share a design; the third-party equation has its own, smaller one, which is the only non-nested pair and therefore the only place a SUR efficiency gain is available (§9).

Turnout is reparameterised as a bounded total times a composition (2026-10-19). Equation (3) models each cell's turnout as an unconstrained log-normal, and the three per-mode turnout equations are fitted independently. Nothing then ties them to each other or to anything physical, and the consequences were severe: statewide turnout read 184% of truth at 20:00 and 107.8% with 100% of the vote counted. The prior mode shares are normalised to sum to one, but only in the prior.

The turnout slot now carries a precinct *total* and a *composition*:

$$\theta_i = \operatorname{logit}\!\left(\frac{T_i}{\kappa\, n_i}\right), \qquad \eta_{im} = \log\frac{p_{im}}{p_{i,\mathrm{ref}}}\ \ (m \neq \mathrm{ref}), \qquad \tau_{im} = T_i\, p_{im}, \quad p_i = \operatorname{softmax}(\eta_i), \tag{4}$$

so $\sum_m \tau_{im} = T_i$ exactly and $T_i < \kappa n_i$ by construction, with n_i the registrants and $\kappa = 1.25$ by default. Both transforms are monotone and the outcomes stay Gaussian, so conjugacy is untouched; the equation count is unchanged at KM, one total plus $M-1$ log-ratios plus the $2M$ share equations, and Σ keeps its dimensions. Two measurements motivate the split. The precinct total is predicted well (corr of logs 0.962, sd of log ratio 0.246) while the mode split is not (sd 0.698 for election day, median ratio 1.87); and a cap only bites on the total, since per-mode caps must exceed each mode's observed maximum share of registrants and those sum to $4.66n_i$ against an actual total of $0.654n_i$. The reference mode is early voting — not because it predicts the others, which it does not ($R^2 = 0.849$ for log(eday) on log(early) + log n against 0.849 for log n alone), but because it reports first and is the best measured.

Each of the $KM = 9$ outcomes gets its own equation, and they are tied together only through their errors:

$$y^k_{im} = x_i'\beta^k_m + \varepsilon^k_{im}, \qquad \boldsymbol{\varepsilon}_i = \left(\varepsilon^{\tau}_{i,\mathrm{mail}},\ \varepsilon^{\mathrm{tp}}_{i,\mathrm{mail}},\ \varepsilon^{s}_{i,\mathrm{mail}},\ \varepsilon^{\tau}_{i,\mathrm{early}},\ \ldots\right)' \ \sim\ \mathcal{N}(\mathbf{0}, \Sigma), \tag{5}$$

with $\Sigma \in \mathbb{R}^{9\times 9}$ and $\boldsymbol{\varepsilon}_i$ independent across i. Note that m is an index, not a multiplicand: β^k_m is a separate p-vector per mode, which is algebraically the same as one pooled regression with mode dummies interacted with every covariate.

This is Zellner's seemingly unrelated regressions (SUR). The equations share no parameters. Σ carries three substantively distinct kinds of dependence:

1. turnout ↔ share *within* a mode;

2. turnout ↔ turnout *across* modes (substitution: heavy early voting implies a lighter election day);
3. share ↔ share *across* modes — which would let an observed mode inform an unobserved mode of the same precinct.

Measured on North Carolina 2024 (1,118 precincts; MEDSL official returns, the only state with complete precinct × mode × party), items 1 and 3 behave very differently. *Within* mode, $\rho(\tau, s)$ is real and sign-varying: +0.204 for mail ($z = 6.9$), +0.052 for early (not significant), −0.140 for election day ($z = -4.7$). *Across* modes, only mail ↔ early is distinguishable from zero (+0.232, $z = 7.9$); mail ↔ election-day is −0.050 ($z = -1.7$) and early ↔ election-day is +0.039 ($z = 1.3$). Cross-mode *turnout* correlation is also ≈ 0 (L2, 5,362 Texas precincts, $|\rho| < 0.2$ in both 2022 and 2024). Σ is therefore imposed **block-diagonal by mode**: the within-mode blocks are kept because they are measured, the cross-mode blocks dropped because they are not.

Texas 2024 now measures the same quantities, and disagrees with North Carolina (2026-10-19). The 2024 Texas scrape yields real precinct × mode × party results, so the blocks that had to be borrowed can be measured directly. On share *residuals* — which is what Σ holds, and not raw share correlations, which are dominated by precinct lean and come out at 0.81–0.94:

pair	TX 2024	NC 2024
mail ↔ early	+0.049 ($t = 2.0$)	+0.232
mail ↔ eday	−0.117 ($t = -4.9$)	−0.050
early ↔ eday	**+0.417** ($t = 22.3$)	+0.039

The pair North Carolina says is negligible is the strongest in Texas, and it is the one that would matter most for the red mirage. Within-mode $\rho(\tau, s)$ also differs: TX gives mail +0.008, early +0.292, election day +0.021. **The NC-borrowed cross-mode block is therefore wrong for Texas in exactly the place it is used.** Until a second state confirms which pattern generalises, the block-diagonal default stands — but it should now be read as "we do not know" rather than "we measured it to be zero".

The mode partisan differential also flips sign between states. Measured on TX 2024 with the specification of §10: mail $a = +0.769$, early $a = +0.087$, election day $a = +0.082$, with two-party Democratic shares by mode of 67.3%, 53.3% and 55.9%. **Texas election day runs** 2.6**pp more Democratic than early voting** — the opposite of North Carolina, and the opposite of the red-mirage pattern the mode machinery was built for. Borrowing NC's δ_m for Texas is accordingly worth nothing: on the 2024 backtest it moves mean |err| from 0.60pp to 0.59pp, and even an oracle δ_m measured on the target election itself only reaches 0.41pp.

3. The prior: a g-prior placed on fitted values

Each equation is given a conjugate prior specified in the units of the outcome rather than on coefficients. Write $\tilde{y}^k_{\cdot m}$ for the vector of prior fitted values across all N precincts in mode m. These come from an external baseline: expected turnout split across modes, the 2024 two-party lean shifted by a mode-specific offset, and a shrunk third-party share,

$$\tilde{\tau}_{im} = \widehat{\text{est}}_i \cdot \pi_{im}, \qquad \tilde{s}_{im} = \text{logit}^{-1}\big(\text{logit}(\text{dem2}_i) + \delta_m\big), \qquad \tilde{\text{tp}}_{im} = \text{tp}_i, \tag{6}$$

where π_{im} are the mode turnout shares and δ_m the mode partisan offsets.

Despite the name, "g-prior" is not shorthand for "Gaussian prior." It is Zellner's specific structure $\beta \sim \mathcal{N}(\beta_0, g\sigma^2(X'X)^{-1})$, in which the prior covariance is proportional to $(X'X)^{-1}$ — the same matrix appearing in the OLS sampling variance — and g is the scalar multiplier. A ridge prior $\mathcal{N}(0, \lambda^{-1}I)$ is

equally Gaussian but is *not* a g-prior, and the difference is exactly what produces items 1 and 3 below. Here $g = 1/w_k$.

The prior is implemented as *data augmentation*. Stack all N precincts as pseudo-observations with response $\tilde{y}^k$ and weight $w_k = n_k/N$, on top of the n_{obs} reported rows with weight one:

$$X_a = \begin{bmatrix} X \\ X_{\text{obs}} \end{bmatrix}, \qquad y_a = \begin{bmatrix} \tilde{y}^k \\ y^k_{\text{obs}} \end{bmatrix}, \qquad W_a = \text{diag}\left(w_k \mathbf{1}_N,\ \mathbf{1}_{n_{\text{obs}}}\right), \tag{7}$$

then solve one weighted least-squares problem. The implied prior on the coefficients is exactly Zellner's g-prior,

$$\beta^k_m \ \sim\ \mathcal{N}\Big(\underbrace{(X'X)^{-1}X'\tilde{y}^k}_{\text{OLS projection of the prior}} \ ,\ \sigma^2_k\big(w_k X'X\big)^{-1}\Big), \qquad \sigma^2_k \sim \text{Scaled-Inv-}\chi^2(\nu_0, s^2_0), \tag{8}$$

with $\nu_0 = 4$ and $s_0 = 0.05$. Three consequences are worth stating plainly.

1. n_k **is an effective sample size in precinct-equivalents.** Because the pseudo-rows live in the same design space and share the same σ, setting $n_\tau = 3$ really does mean "three precincts' worth of information." The prior's *relative* weight then decays as $n_k/(n_k + n_{\text{obs}})$ with no intervention, which is why n_k is held fixed through the night rather than tuned downward as returns arrive. Shrinking it as well would double-discount the prior and overstate confidence in early partial data.
2. **The prior mean is a projection, so it can be lossy.** Only the component of $\tilde{y}^k$ lying in $\text{col}(X)$ survives (8). Including the baseline itself as a covariate makes the projection exact; omitting it discards whatever structure the covariates cannot express.
3. **The prior precision is proportional to $X'X$.** It is therefore tightest in the directions the design is already most informative about, and vanishes in near-null directions. Unlike a ridge penalty λI, no choice of n_k improves the conditioning of the design; the prior fixes $n_{\text{obs}} < p$ early in the night but not collinearity.

4. The county random effect

Equation (5) as written has $\boldsymbol{\varepsilon}_i$ independent across precincts. That assumption is false, and it is the single most consequential error the model can make. For a sum of N precinct-level terms with mean pairwise correlation $\bar{\rho}$,

$$\text{Var}\left(\sum_{i=1}^{N} X_i\right) = N\sigma^2\big(1 + (N-1)\bar{\rho}\big), \tag{9}$$

so the standard deviation is inflated by $\sqrt{1 + (N-1)\bar{\rho}}$ relative to independence. With $N \approx 4{,}500$ a $\bar{\rho}$ of one *thousandth* roughly doubles a statewide interval — while every other dependence examined in this model moves it by under one percent.

4.1. What the data say

Residuals cluster by county, and not marginally. Measuring the county intraclass correlation of a top-of-ticket statewide share residual (regressed on another statewide office on the same ballot, vote-weighted, plus log size), on MEDSL official precinct returns:

Three things follow. The effect is *universal* — every state-year shows it. The inflation is *stable* at 2.4× to 5.1×, so any precinct model treating precincts as independent is overconfident by roughly a factor of three essentially everywhere. And midterm ICCs generally exceed presidential ones, so for a 2026 forecast the 2022 column is the relevant analogue.

	county ICC		implied sd inflation	
state	2022	2024	2022	2024
TX	**0.269**	**0.196**	3.07×	2.72×
NC	—	0.430	—	3.47×
GA	0.295	—	2.37×	—
MI	0.381	0.149	4.61×	2.90×
PA	0.091	0.091	3.38×	3.52×
WI	0.114	0.144	2.37×	2.66×
AZ	0.089	0.227	3.19×	5.12×
OH	0.193	0.131	4.31×	3.70×
NV	0.106	0.142	2.81×	3.22×

Table 2: County ICC of the share residual, nine states and two cycles. The ICC ranges widely but the inflation is tightly clustered, because inflation depends on ICC × precincts-per-county and states trade off between the two.

The defaults are the Texas measurements, avoiding cross-state borrowing on the parameter that matters most: $\text{ICC}_s = 0.269$ (MEDSL TX 2022, governor on attorney general, 7,529 precincts in 233 counties) and $\text{ICC}_\tau = 0.200$ (L2 Texas, log(`vote22`) on registration, demographics and 2018 turnout, 5,433 precincts in 32 counties).

A coarse stratification will not substitute. Regressing the NC county effects on county attributes gives $R^2 = 0.005$ for lean, 0.281 for log size, 0.396 for both with cubic terms, and 0.034 for a six-level stratification on lean. Roughly 60% of the county effect is county-*idiosyncratic*, which is why neither existing pipeline captures it: the NYC model has no grouping term at all, and the TX primary model's (1 | `stratum_demo`) and (1 | `geostratum`) have five and six levels over 254 counties.

4.2. Implementation: a ridge-equivalent effect that preserves conjugacy

Let C be the $N \times J$ matrix of county indicators and $u \in \mathbb{R}^J$ the county effects. The model becomes

$$y^k_{im} = x_i'\beta^k_m + u^k_{m,c(i)} + \varepsilon^k_{im}, \qquad u^k_{m,c} \sim \mathcal{N}\left(0,\ \sigma^2_k/\lambda_k\right), \qquad \lambda_k = \frac{1 - \text{ICC}_k}{\text{ICC}_k}. \tag{10}$$

Fitting this with `lme4` would forfeit conjugacy and the closed-form draws of (15). Instead, note that a Gaussian random effect is exactly a ridge penalty on the indicator columns, and that a ridge penalty is representable as pseudo-rows. The augmented system stacks three blocks and is solved by one weighted QR:

$$\underbrace{\begin{bmatrix} X & \mathbf{0} \\ X_{\text{obs}} & C_{\text{obs}} \\ \mathbf{0} & I_J \end{bmatrix}}_{\text{design}}, \qquad \begin{bmatrix} \tilde{y}^k \\ y^k_{\text{obs}} \\ \mathbf{0} \end{bmatrix}, \qquad W = \text{diag}\left(\tfrac{n_k}{N}\mathbf{1}_N,\ w_{\text{obs}},\ \lambda_k \mathbf{1}_J\right). \tag{11}$$

The first block is the g-prior of §3, with the county columns zeroed so that it governs β only; the second is the reported data, weighted by counted ballots; the third shrinks u toward zero with weight λ_k. Conjugacy, the closed-form draws, and the precinct-equivalents reading of n_k all survive.

The behaviour that produces the widening is at the edge: a county with *no* reported precincts has u_c shrunk to zero but retains the full random-effect variance σ^2_k/λ_k in the posterior draws. Unreported counties are therefore genuinely uncertain rather than silently assumed average.

4.3. Validation

Coverage over 14 replications with a fresh truth each (county component at the measured ICCs), single stage at 70% reported, nominal 95%:

	coverage	mean width	mean |error|
county effect off	8/14 = 57%	0.577pp	0.326pp
county effect on	13/14 = 93%	1.331pp	0.232pp
off, at the TX ICCs	11/14 = 79%	0.526pp	0.218pp
on, at the TX ICCs	13/14 = 93%	1.012pp	0.235pp

Table 3: The uncorrected model's nominal 95% intervals cover 57–79% of the time. With the county effect, 93% — indistinguishable from nominal given $n = 14$ (SE on coverage ≈ 5.8pp).

Two caveats. The synthetic truth is generated with the same ICCs the model assumes, so this establishes that the machinery is calibrated *when the ICC is correct*, not that 0.269 is correct for 2026. And $n = 14$ is thin: the 93% figures are consistent with roughly 82–100%, while the 57% and 79% figures are far enough from nominal to survive that noise.

One component remains unmeasured. Equation (10) captures the *county* level. A *statewide* component — a uniform swing the covariates miss — is not identifiable from a single election's residuals, because the fitted intercept absorbs it by construction. The inflation factors in Table 2 are therefore lower bounds, and recovering the statewide term needs either multi-election backtesting or an explicit prior, which is what the systematic term in §8 is.

5. Posterior and posterior draws

With (7) the posterior is closed-form. Let R be the upper-triangular factor from the QR decomposition of $W_a^{1/2}X_a$, so $R'R = X_a'W_aX_a$. Then

$$\hat{\beta} = \left(X_a'W_aX_a\right)^{-1}X_a'W_ay_a, \qquad \hat{\sigma}^2 = \frac{\left\|W_a^{1/2}\left(y_a - X_a\hat{\beta}\right)\right\|^2}{n_{\text{obs}} - p}, \tag{12}$$

and the variance posterior is updated conjugately,

$$\nu_\star = \nu_0 + N + n_{\text{obs}}, \qquad s_\star^2 = \frac{\nu_0 s_0^2 + \left(N + n_{\text{obs}}\right)\hat{\sigma}^2}{\nu_\star}. \tag{13}$$

Corrected 2026-10-19 — the previous version of this was a bug. Earlier drafts described the residual sum of squares as running over the *augmented* rows while the degrees of freedom used only the reported rows, and defended that as "crude but directionally correct self-protection". It is not defensible: the numerator and the denominator then describe different row sets, and the error is unbounded in both directions. With n_{obs} small relative to N and J, the g-prior and ridge blocks dominate the numerator and σ explodes — the election-day equation fitted on one county's 15 cells returned $\sigma = 2.19$ on the log-turnout scale, so $\exp(\mu + \varepsilon)$ produced statewide turnout draws of order 10^{12}. In the other direction, setting $\nu_\star = \nu_0 + N + n_{\text{obs}}$ counts the pseudo-rows as data, which makes σ effectively *known* and strips its uncertainty out of every predictive interval; late-night widths collapsed to 0.02pp while the point estimate was off by 0.75pp. Both symptoms came from the same mis-specification.

Residual variance is now computed on the data rows alone, with the degrees of freedom consumed measured exactly rather than assumed:

$$\widehat{\mathrm{edf}} = \mathrm{tr}\big(H_{\mathcal{D}\mathcal{D}}\big) = \sum_{r\in\mathcal{D}} \|Q_{r\cdot}\|^2, \qquad \nu = n_{\mathrm{obs}} - \widehat{\mathrm{edf}}, \qquad \hat\sigma^2 = \frac{\sum_{r\in\mathcal{D}} \big(W_a^{1/2}(y_a - X_a\hat\beta)\big)_r^2}{\nu}, \tag{14}$$

where $\mathcal{D}$ indexes the reported rows and Q is the thin Q factor of $W_a^{1/2}X_a$. The trace matters because the county block is penalised: it consumes well under J degrees of freedom, and when only a few county levels are identified by data at all, assuming otherwise is badly wrong. The conjugate update then runs on the data degrees of freedom, $\nu_\star = \nu_0 + \nu$ and $s_\star^2 = (\nu_0 s_0^2 + \nu\hat\sigma^2)/\nu_\star$.

One consequence is worth stating plainly, because it cost a day to understand. The old inflated σ was accidentally supplying predictive variance for *unobserved* cells, which the model has no other source for. Removing the bug removed that too, and synthetic precinct coverage fell from 0.98 to 0.65. The missing component is real and is treated in §8; the old formula was a badly-scaled proxy for it, not a substitute.

Draws are then taken directly:

$$\sigma^{(d)} = s_\star\sqrt{\nu_\star/\chi^2_{\nu_\star}}, \qquad \beta^{(d)} = \hat\beta + R^{-1}z^{(d)}, \qquad z^{(d)} \sim \mathcal{N}\big(\mathbf{0},\ \sigma^{(d)2}I_p\big), \tag{15}$$

which is a multivariate t for β marginally. We take $D = 500$ draws.

Why Gaussian rather than Poisson counts

A Poisson or negative-binomial model of candidate counts is the natural alternative and is what the Texas primary pipeline uses. The Poisson log-likelihood $\sum_i\{v_i x_i'\beta - e^{x_i'\beta}\}$ is not quadratic in β, so no finite-dimensional conjugate family exists — Gamma conjugacy applies to a free rate λ_i, and imposing $\lambda_i = e^{x_i'\beta}$ with β shared across precincts destroys it. One must then optimise or sample, which returns a point estimate rather than a posterior, and intervals have to be manufactured some other way. Keeping everything Gaussian preserves (15) in closed form, which is what makes refitting many races every few minutes feasible.

6. Σ and conditioning on reported modes

Σ is treated as an *input*, estimated once from a past election in which every mode eventually reported, as the complete-case covariance of the $N \times 9$ residual matrix (with eigenvalue clipping to restore positive-definiteness).

Σ cannot be estimated from tonight's returns

Modes report in sequence — mail, then early, then election day — so the number of precincts with *all* modes counted is near zero until counting is essentially over. The share ↔ share block linking mail to election day, which is precisely the most useful one, is never identified from the current night's data. It has to come from history. An earlier implementation estimated Σ by complete cases from the live returns and silently fell back to a diagonal matrix at every reporting stage, making the whole mechanism inert.

Partition precinct i's error vector into the components belonging to reported modes (o) and unreported modes (u). Prediction for the unreported cells then uses the conditional normal rather than the marginal:

$$\boldsymbol{\varepsilon}_{iu} \mid \boldsymbol{\varepsilon}_{io} \ \sim\ \mathcal{N}\Big(\Sigma_{uo}\Sigma_{oo}^{-1}\boldsymbol{\varepsilon}_{io},\ \ \Sigma_{uu} - \Sigma_{uo}\Sigma_{oo}^{-1}\Sigma_{ou}\Big), \qquad \varepsilon_{im}^{k(d)} = y_{im}^{k} - x_i'\beta_m^{k(d)}. \tag{16}$$

Concretely: if a precinct's mail vote lands three points more Democratic than predicted, its election-day prediction shifts by roughly $\rho \cdot 3 \cdot (\sigma_{\text{eday}}/\sigma_{\text{mail}})$ points and its variance shrinks by the Schur complement. Without (16), observing one mode tells you nothing about another mode of the same precinct beyond what the covariates already said.

Since the observed/unobserved pattern is shared by many precincts, (16) is computed once per distinct pattern — at most $2^M = 8$ of them.

7. Prediction and aggregation

Within each draw d: form the latent outcomes from (5) using $\beta^{(d)}$ and the residual draw from (16), invert the transforms of (3) and (4), combine each cell's counted votes with its predicted remainder, and only then aggregate via (2):

Partial counts enter as a mixture, not a substitution (2026-10-19). A reported cell used to have its counted votes substituted as if final. Real snapshots are cumulative *partial* counts: measured on Texas 2024, when an election-day cell first shows votes the median cell has 5.4% of its final count in, and mail sits at 66% of final at 19:00. Treating those as final understated statewide turnout by 12.5pp at 22:30 and biased the share toward the under-counted mode's opponent. The binary alternative — drop the cell until it is "complete" — is worse, because it discards a cell that is 99% counted. Writing C_{im} for votes counted so far in the cell and $C_i = \sum_m C_{im}$,

$$T_i^{(d)} \;\geq\; C_i, \qquad r_{im}^{(d)} \;=\; \left(T_i^{(d)} - C_i\right) \frac{\left[T_i^{(d)} p_{im}^{(d)} - C_{im}\right]_+}{\sum_{m'} \left[T_i^{(d)} p_{im'}^{(d)} - C_{im'}\right]_+}, \qquad v_{im}^{\text{dem}(d)} = C_{im}^{\text{dem}} + r_{im}^{(d)} \left(1 - \text{tp}_{im}^{(d)}\right) s_{im}^{(d)}, \tag{17}$$

and analogously for rep and oth. A counted-out cell has $r \to 0$ and pins itself; a barely-started cell is mostly predicted. This needs **no completeness classifier**, which matters because in production the `stage` field reaches "complete" days late, uses a different vocabulary per state, is downgraded by our own loader, and has been observed moving *backwards* (`pct_rep` $0.997 \to 0.269 \to 0.364$). The floor acts on the precinct total rather than per cell: flooring cells independently lets $\sum_m \tau_{im}$ exceed T_i again and cost 8pp of turnout accuracy at 99% counted when it was first written that way.

$$S_{\text{dem}}^{(d)} = \frac{\sum_{i,m} \tau_{im}^{(d)} \left(1 - \text{tp}_{im}^{(d)}\right) s_{im}^{(d)}}{\sum_{i,m} \tau_{im}^{(d)}}, \qquad S_{\text{oth}}^{(d)} = \frac{\sum_{i,m} \tau_{im}^{(d)} \text{tp}_{im}^{(d)}}{\sum_{i,m} \tau_{im}^{(d)}}, \tag{18}$$

and $S_{\text{rep}}^{(d)}$ analogously. The headline quantities are these shares of all votes; they satisfy $S_{\text{dem}} + S_{\text{rep}} + S_{\text{oth}} = 1$ identically, since the numerators sum to $\sum \tau$. The two-party share $S_{\text{dem}}/(S_{\text{dem}} + S_{\text{rep}})$ is reported alongside, because it is what decides the race.

The ordering is not cosmetic. Because exp, logit^{-1} and the ratio in (18) are all nonlinear, aggregating within each draw and taking quantiles afterwards is not the same as — and is the correct alternative to — forming per-cell intervals and combining them.

8. Intervals

Two intervals are reported for the headline quantity, and they answer different questions.

The **credible interval** summarises the posterior draws and is a fixed-sample statement: given the model, there is $1 - \alpha$ posterior probability that the quantity lies in the interval, *for one look at the data*.

The **confidence sequence** is anytime-valid. It is the set of values whose posterior-to-prior density ratio has not yet crossed $1/\alpha$,

$$\mathrm{CS}_\alpha = \left\{ y : \frac{p_{\text{post}}(y)}{p_{\text{prior}}(y)} \le \frac{1}{\alpha} \right\}. \tag{19}$$

That ratio is a Bayes factor against the hypothesis that the truth is y; by Ville's inequality it is a non-negative martingale with unit expectation under the truth, so it exceeds $1/\alpha$ with probability at most α *at any stopping time*. Coverage therefore holds uniformly across the whole night. This matters because the model refits every few minutes: with on the order of a hundred looks, a freshly computed 95% credible interval has nowhere near 95% joint coverage, while (19) does by construction.

The price is width. For Gaussian prior and posterior with scales $\sigma_0 > \sigma_1$, the confidence sequence runs roughly 1.2–1.6 times wider than the credible interval, the ratio shrinking toward one as $\alpha \to 0$ because both widths grow like $\sqrt{\log(1/\alpha)}$. When the posterior has not concentrated relative to the prior the ratio in (19) can never reach $1/\alpha$ and the interval is $(-\infty, \infty)$ — which is the honest answer early in the night, and something a credible interval will never tell you.

Note that (19) requires the prior predictive as its reference measure, and that this must include a *systematic* component. A prior built only from idiosyncratic per-cell noise averages out at $1/\sqrt{NM}$ and comes out tighter than the posterior, which makes the governing quadratic unbounded and returns an infinite interval at every stage. Real prior uncertainty is about the statewide swing, which is common to all precincts and does not average away. That scale is now measured rather than assumed: across 37 ordered contest pairs in 9 states, the statewide two-party share moves 0.22–0.26 logit units between a prior contest and its target, against the $\sigma/\sqrt{n_{\text{prior}}} \approx 0.098$ the g-prior implicitly assumed.

A wide prior on the unreported set (2026-10-19). The cells still to be counted within a (county, mode) differ from that county's already-counted ones by more than any existing term can express. Measured on Texas 2024, the logit gap between the remaining and the counted share has *mean* ≈ 0 (-0.10 to $+0.13$ across stages) and sd 0.26–0.45, stable under leave-one-county-out (0.337–0.395 against a full-sample 0.373). Mean zero with large stable dispersion is a variance component, not a predictable offset — and the predictor route is separately refuted, since within (county, mode) pre-election covariates explain only 4.6% of election-day lateness and lateness → share is $t = -1.4$. The model therefore adds one shared mean-zero offset per (county, mode) draw to the share of cells that are not counted out, ε^s_{im} += $u_{c(i),m}$ with $u \sim \mathcal{N}(0, \sigma_u^2)$, $\sigma_u = 0.25$. Being mean-zero it must not move the point estimate, only the width — which is the sharpest available check that it is wired correctly. Nothing already in the model covers it: per-cell residuals average over the unreported set at $1/\sqrt{k}$, giving ≈ 0.02 on the logit scale, two orders below the measured 0.3; and u_c for a (county, mode) is pinned by that county's *reported* cells, so it cannot express a reported-versus-unreported difference.

9. Assumptions and known gaps

Reporting is ignorable given covariates — now partly measured, and it largely holds. A cell enters the likelihood when it has reported, and early reporters are not a random sample: on Texas 2024 the early vote counted at 18:45 reads 62.95% two-party Democratic against an early-vote final of 53.29%, a raw selection of nearly ten points *within a single mode*. The question is whether the covariates absorb it. They largely do. `dem2` alone explains $R^2 = 0.938$ of early-vote logit share and the full design 0.963; after fitting, county-mean residuals do *not* significantly track when a county reported ($\rho = 0.268$, $t = 1.04$, $p = 0.32$, $n = 16$, weighted residual sd 0.057 logit). In the backtest the model removes over half the raw skew: the counted sample is +1.76pp at 19:30 and the estimate lands at +0.79pp.

Two cautions. Three of the sixteen counties post early-vote totals *before* poll close and all three carry negative residuals, which drags that correlation toward zero, so a weak effect may exist that $n = 16$ cannot resolve. And the residual +0.79pp is better explained by the prior's level than by geography: `dem2` is 52.22% vote-weighted against an actual two-party 54.33%, so the prior pulls down while the skewed counted sample pulls up. That is a prior-level problem, and loosening the level prior is *not* the fix — doing so lets the biased early sample pull harder and degrades the point estimate (max |err| 1.40 → 2.23pp).

Errors are independent across precincts. This is the most consequential remaining gap. Summing N precincts with independent errors gives aggregate variance $O(N)$; with correlated errors it is $O(N^2)$, and that difference is what sets the true width of a statewide band. Earlier drafts of this memo reported that cross-mode conditioning improves cell-level share prediction by 21% in RMSE and statewide standard deviation by 10%. **Both figures were artefacts and are withdrawn.** The first came from a synthetic generator that planted $\rho \approx 0.88$ across modes, so it measured only that exploiting an assumption helps; the real value is ≈ 0 for the mail ↔ election-day pair. The second compared two Σ matrices that differed in their *diagonal* as well, so it measured an unreconciled variance estimate rather than correlation structure. Holding the diagonal fixed and varying only the off-diagonals moves the statewide standard deviation by 0.09%; propagating the measured NC correlations gives 0.09% statewide, 0.21% for a 500-precinct district and 0.86% for a 30-precinct one. Correlation structure is thus second-order at every aggregate scale, because the residual term contributes only $O(1/\sqrt{k})$ of total variance while coefficient uncertainty moves every precinct together and does not average down. A geostratum random effect or explicit spatial term remains the highest-value addition.

Partial reporting is treated as all-or-nothing. A cell counts as observed at `STAGE_DONE` and is otherwise unobserved. In reality a half-counted cell is asymmetric between the equations: a *weighted* observation for the share equations, but a *censored* one for turnout, since the count so far is a lower bound on τ_{im}. Treating it as observed would bias turnout downward across nearly every cell early in the night.

The mode partisan offsets δ_m are not yet calibrated, and must be lean-dependent rather than constant. Measured on NC 2024, mail runs +0.660 logit above its precinct's overall two-party share on average, with a slope of −0.297 on the precinct's own lean ($t = -20.8$); early and election day are near zero on average but also lean-dependent ($t = 17.5$ and -11.8). A constant offset is therefore misspecified, not approximately right. The *likelihood* already accommodates this, since each mode has its own β_m^s with the partisan baseline in the design; it was the *prior* that was wrong, and it is now $\delta_{im} = a_m + b_m \,\mathrm{logit}(\mathrm{dem2}_i)$. Supplying a correct lean-dependent offset cuts the error at 10% reported from 1.07 to 0.17 percentage points in the trial. The coefficients are left at zero for Texas because NC has no-excuse mail voting while Texas restricts it to voters 65+, disabled or absent, so the magnitudes should not be transferred; NC establishes that a nonzero, lean-dependent offset is needed, not its size. Voter-file sources give mode turnout composition — for Texas the 2022 midterm split is mail 4.2%, early 63.8%, election day 32.0%, against 3.5/76.4/20.2 in 2024, so a presidential-year analogue would understate election day by about ten points — but they record how people voted, never for whom. The offsets therefore require either precinct × mode × party results, or ecological inference crossing precinct mode composition with precinct partisan outcomes. Until then the trial reproduces the full red-mirage bias on purpose, and does so visibly.

The SUR system is not fully exploited. Coefficients are estimated equation-by-equation and Σ is used only at prediction time. Where designs coincide this costs nothing: with identical regressors the GLS estimator collapses algebraically to equation-by-equation least squares. Since the turnout and share

equations currently share a design, the only available gain comes through the third-party equation's distinct design and is worth about 1%. Differentiating the turnout and share designs would raise it to roughly 10%, but that requires genuine exclusion restrictions and an iterative (Gibbs/feasible-GLS) fit, forfeiting the closed form of (15). Given the cost of refitting many races continuously, that trade is currently unattractive.

10. Implementation

Per refit, per race: nine QR decompositions of the augmented design (7), plus at most eight small solves for (16), each inverting a Σ_{oo} of order at most nine with one Cholesky of the Schur complement. Nothing iterates and nothing has to converge.

Equation	Quantity	Code
(1), (3)	observed outcomes, corrected and raw	`fit_stage` step 4
(5), (7), (8)	nine conjugate fits	`fit_stage` step 4a
(15)	coefficient draws	`draw_beta`
Σ	source and repair	`fit_stage` 4b, `calibrate_sigma`
(16)	conditioning by observation pattern	`fit_stage` 4c
(2), (18)	invert, substitute, aggregate	`fit_stage` 4d
(19)	anytime-valid interval	`conf_seq`, `prior_draws`

The conjugate regression, the credible interval and the confidence sequence come from the `enightmodels` package (`lm_bayes`, `cred_int`, `conf_seq`); everything else is in `tx_gov_trial.R`. The trial runs on synthetic covariates and synthetic returns when the real inputs are absent, so the plumbing is testable before data arrives; synthetic mode is labelled loudly at run time.

11. Validation: the Texas 2024 real reporting cadence

Earlier drafts closed with a request to backtest on a real reporting order rather than a synthetic one. That has now been done, and it changed the model.

Why the synthetic orders were not enough. Two reporting orders had been tried, both invented. A representative order gave 8/8 coverage and $|\text{err}| \le 0.36$pp — too benign. An adversarial order in which small rural counties finish first gave 2/7 coverage and errors to −16.5pp. The real Texas curve is neither: all sixteen counties report *something* within minutes of poll close, so geographic coverage is nearly complete from the start, but the first batches are ≈ 97% early vote. Texas posts 47% of its entire vote in the first batch after poll close, and that batch reads 8.9pp more Democratic than the final result. Neither invented order produced that, and the defects it exposed were invisible to both.

Construction. The panel is rebuilt from 249 raw county scrapes into 98,201 (precinct, mode, timestamp) cells over 16 counties and 4 modes, reproducing an independent reference exactly (identical key set, zero vote differences). Snapshots are cumulative *within* a county and counties are scraped independently, so every statewide figure uses an as-of join — each county's latest snapshot at or before t. Truth is the final canvass over the same precinct universe: 2,417 precincts, 52.950% Democratic of all votes, 54.327% two-party. All predictors predate November 2024: `dem2` and the third-party prior from the 2020 Senate race, expected turnout from 2020, demographics and turnout history from 2018 and 2022.

Result. Against a naive count-what-is-in benchmark that is +8.94pp wrong at 18:45, the model holds ≤ 1.4pp at every stage, with mean |err| of 0.42–0.49pp across configurations, and the interval collapses

as the count completes.

as-of	% votes	naive	model	width
18:45	16.2	+8.94	−1.40	6.97
19:15	53.7	−0.96	+0.30	3.90
21:00	78.2	−0.65	+0.16	2.45
23:00	89.2	−0.82	−0.17	2.50
03:00	99.2	−0.09	−0.06	0.35
final	100.0	0.00	−0.03	0.44

What it found. Three defects that neither synthetic order could reach, each documented in the section it belongs to: the variance mis-specification of §5; partial reporting, §7, which the all-or-nothing synthetic generator cannot produce at all; and unbounded per-mode turnout, §2. Turnout at 20:00 improved from 184% of truth to 88% once the composition was imposed, and from 107.8% to 103.7% at 100% counted.

What it does not establish. The fifteen stages are nested snapshots of one election with one truth, so a coverage count across them is not a coverage *rate* — it behaves closer to a single observation, and varying the random seed tests Monte Carlo stability rather than calibration. The honest claim is narrower than the numbers invite: the fixes removed clear evidence of *under*-coverage, and the point-estimate curve is good. Demonstrating calibration needs independent replications — other states, other elections — which the multi-state panel work exists to supply. The synthetic path can no longer serve as that check either: its generator reveals cells all-or-nothing, so it cannot exercise either partial-reporting fix, and its coverage figures are uninterpretable until it emits partial counts.

12. Next steps

Items 1, 3 and 5 of the previous list are done or superseded; what remains is reordered by what the Texas 2024 backtest showed actually binds.

1. **Replicate on a second state-election.** This is now the binding item: every number in §11 comes from one election, and calibration cannot be established from it. North Carolina is the best candidate — full-state coverage, 100 counties, mode × party available — and its crosswalk is 98.6% populated, so it is not blocked.
2. **Separate level uncertainty from level shrinkage.** The measured statewide swing (0.22–0.26 logit) is the right scale for interval *width* and the wrong one for shrinkage: applying it to the prior degrades the point estimate, because a looser prior lets the biased early sample pull the level. The likely shape is a mean-zero statewide offset on the predictive draw, as in §8, rather than a weaker posterior pull. Implemented but defaulted off pending that distinction.
3. **An atom at the bound for censored turnout.** Right-censoring is implemented (EM, conjugacy preserved) and currently *off*: it fixes early turnout (184% → 99%) and breaks late (100% → 124%), because pure Tobit gives a finished cell zero probability of sitting exactly at its bound when $\tau = C$ exactly is the common case. The fix is a mixture likelihood with a discretised probability atom, not a different threshold.
4. **Cross-precinct correlation** beyond the county effect — still the term that governs statewide width, and still absent.
5. **Make the synthetic generator emit partial cumulative counts** and a completeness flag, so it can once again serve as a regression check.

6. Replace the third-party prior with observed past third-party share once precinct identifiers are reconciled across vintages.